

# predict\_spatial\_resolution\_grid\_age.R

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predict\_spatial\_resolution\_grid\_age  
*Predict Spatial-Resolution Grid for One Age*

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## Description

Builds abiotic and optional spatial predictors for one age slice and returns long-format taxon occurrence probabilities.

## Usage

```
predict_spatial_resolution_grid_age(  
  prediction_inputs,  
  data_grid,  
  data_grid_coords_projected,  
  age,  
  abiotic_variables,  
  x_lim,  
  y_lim,  
  cache_dir,  
  spatial_mode = "spatiotemporal",  
  climate_fn = NULL,  
  spatial_interpolate_fn = NULL,  
  spatial_basis_project_fn = NULL,  
  spatiotemporal_interpolate_fn = NULL,  
  predict_fn = NULL  
)
```

## Arguments

prediction\_inputs  
Named list returned by [load\_spatial\_resolution\_prediction\_inputs()].

**data\_grid** Prediction grid with 'grid\_id', 'coord\_long', and 'coord\_lat'.  
**data\_grid\_coords\_projected** Projected prediction grid coordinates with row names 'grid\_<id>'.  
**age** Numeric scalar age in years before present.  
**abiotic\_variables** Character vector of CHELSA variables to extract.  
**x\_lim** Numeric longitude range.  
**y\_lim** Numeric latitude range.  
**cache\_dir** Existing CHELSA cache directory.  
**spatial\_mode** Spatial predictor mode: "spatial", "spatiotemporal", or "none".  
**climate\_fn** Climate extraction function. If 'NULL' (default), [extract\_prediction\_climate()] is used.  
**spatial\_interpolate\_fn** Spatial interpolation function. If 'NULL' (default), [interpolate\_mev\_to\_grid()] is used.  
**spatial\_basis\_project\_fn** Reusable-basis projection function. If 'NULL' (default), [project\_spatial\_mev\_basis()] is used when basis state is available.  
**spatiotemporal\_interpolate\_fn** Spatiotemporal interpolation function. If 'NULL' (default), [interpolate\_st\_mev\_to\_grid()] is used.  
**predict\_fn** Prediction function. If 'NULL' (default), the function resolves 'sjSDM::predict.sjSDM'.

### Value

Long tibble with predicted probabilities by grid cell, age, and taxon.

### Examples

```

## Not run:
predict_spatial_resolution_grid_age(
  prediction_inputs = prediction_inputs,
  data_grid = data_grid,
  data_grid_coords_projected = data_grid_coords_projected,
  age = 0,
  abiotic_variables = c("bio1", "bio12"),
  x_lim = c(-10, 40),
  y_lim = c(35, 70),
  cache_dir = "Data/Temp/chelsa/prediction"
)

## End(Not run)

```

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